



Question 1

4. Assume that production $T \rightarrow T * F$, provided that T , $*$, and F are grammar symbols, is a final item of a given LR(0) item. How many grammar symbols will be popped from the parser stack?

Select one:

- a. 2 grammar symbols
- b. 6 grammar symbols
- c. 1 grammar symbols
- d. 3 grammar symbols

Feedback

The correct answer is: 6 grammar symbols

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Question 2

19. Which of the following is not a step in elimination of states procedure?

Select one:

- a. Unifying all the final states into one using e-transitions
- b. Unify single transitions to multi transitions that contains union of input
- c. Remove states until there is only starting and accepting states
- d. Get the resulting regular expression by direct calculation

Feedback

The correct answer is: Unify single transitions to multi transitions that contains union of input

Question 3

54. What is the preferred way for adding a background color in HTML?

Select one:

- a. <background>yellow</background>;
- b. <body background="yellow">;
- c. <background color="yellow">text<background>;
- d. < body style="background-color:yellow">;

Feedback

The correct answer is: < body style="background-color:yellow">;

Question 4

66. While sorting the elements of the array in ascending order using quicksort, the worst-case time complexity will happen if the elements in the array have:

Select one:

- a. The same values or are already arranged in ascending order.
- b. Different values or arbitrary arrangement.
- c. Arbitrary arrangement.
- d. Different values or are already arranged in descending order.

Feedback

The correct answer is: The same values or are already arranged in ascending order.

Question 5

13. _____ refers to the objective of an artificial machine.

Select one:

- a. Input
- b. Initial state
- c. Goal
- d. Intermediate state

Feedback

The correct answer is: Goal

Question 6

40. Which statement correctly differentiates functions and procedures?

Select one:

- a. A function may return one or more values to the calling environment, whereas a procedure must return a single value to its calling environment.
- b. They only have syntax difference otherwise similar
- c. A function can be called only as part of a SQL statement, whereas a procedure can be called only as a PL/SQL statement.
- d. A function must return a value to the calling environment, whereas a procedure can return zero or more values to its calling environment.

Feedback

The correct answer is: A function must return a value to the calling environment, whereas a procedure can return zero or more values to its calling environment.

Question 7

88. A new method of dealing with applets that works by dividing the virtual address space up into equal-size regions is called _____ .

Select one:

- a. Interpreter
- b. Trusted applet
- c. Sandboxes
- d. Untrusted applet

Feedback

The correct answer is: Sandboxes

Question 8

60. Which of the following is a correct order of activities in system development life cycle?

Select one:

- a. Requirement Collection -> Analysis -> Coding -> Design -> Testing -> Deployment -> Maintenance
- b. Requirement Collection -> Analysis -> Design -> Coding -> Testing -> Maintenance -> Deployment
- c. Requirement Collection -> Design -> Analysis -> Coding -> Testing -> Maintenance -> Deployment
- d. Requirement Collection -> Analysis -> Design -> Coding -> Testing -> Deployment -> Maintenance

Feedback

The correct answer is: Requirement Collection -> Analysis -> Design -> Coding -> Testing -> Deployment -> Maintenance

Question 9

65. The UML diagram that depicts the interaction between entities such as objects, classes, user interfaces etc participated in an execution of a certain use case is known as

Select one:

- a. State diagram
- b. Sequence diagram
- c. Use case modeling
- d. Class diagram

Feedback

The correct answer is: Use case modeling

Question 10

80. Which tool is used for managing and controlling network access and security policies on a Windows Server?

Select one:

- a. Group policy management
- b. Network load balancing manager
- c. Container
- d. Organizational Unit

Feedback

The correct answer is: Organizational Unit

Question 11

70. Which of the following is true about spanning trees?

Select one:

- a. A spanning tree will have $n-1$ vertices, if the original graph has n -number of vertices.
- b. A spanning tree will have $n+1$ edge, if the original graph has n -number of vertices.
- c. A spanning tree will have $n-1$ edges, if the original graph has n -number of vertices.
- d. A spanning tree will have n edges, if the original graph has n -number of vertices.

Feedback

The correct answer is: A spanning tree will have $n-1$ edges, if the original graph has n -number of vertices.

Question 12

45. Which one of the following RAID level offers optimal performance and reliability?

Select one:

- a. 0
- b. Any RAID level
- c. 1
- d. 0+1

Feedback

The correct answer is: 0+1

Question 13

9. Which of the following IPv4 address belongs to class B address?

Select one:

- a. 192.1668.1.248
- b. 172.12.12.48
- c. 224.1.120.29
- d. 121.12.12.8

Feedback

The correct answer is: 172.12.12.48

Question 14

6. Which of the following is not essential component of a data communications system?

Select one:

- a. Network Topology
- b. Sender
- c. Transmission media
- d. Protocol

Feedback

The correct answer is: Network Topology

Question 15

79. Which component of Active Directory provides a searchable catalog of objects across multiple domains in a forest?

Select one:

- a. Domain
- b. Global Catalog
- c. Sites
- d. Domain controller

Feedback

The correct answer is: Global Catalog

Question 16

39. Which one of the following statement is TRUE about database normalization?

Select one:

- a. A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R
- b. Every relation in BCNF is also in 3NF
- c. Every relation in 3NF is also in BCNF
- d. No relation be both in BCNF & 3NF

Feedback

The correct answer is: Every relation in BCNF is also in 3NF

Question 17

The relation book (title, price) contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list?

```
SELECT title
```

```
FROM book as B
```

```
WHERE (SELECT count(*)
```

```
FROM book as T
```

```
WHERE T.price>B.price) < 5
```

Select one:

- a. Title of the fifth most inexpensive book
- b. Titles of the four most expensive books
- c. Titles of the five most expensive books

d. Title of the fifth most expensive book

Feedback

Your answer is correct.

The correct answer is: Titles of the five most expensive books

Question 18

14. Which one of the following is a correct representation of the fact "Abebe is father of Natan" using first order logic in prolog?

Select one:

- a. Father(abebe).
- b. black(abebe).
- c. Father(natan).
- d. father(abebe, natan).

Feedback

The correct answer is: father(abebe, natan).

Question 19

83. Which command-line tool is used to display the current TCP/IP network configuration on Windows?

Select one:

- a. Tracert
- b. Nslookup
- c. Ping
- d. Ipconfig

Feedback

The correct answer is: Ipconfig

Question 20

76. 1. A Double-ended queue supports operations such as adding and removing items from both the sides of the queue. They support four operations like addFront(adding item to top of the queue), addRear(adding item to the bottom of the queue), removeFront(removing item from the top of the queue) and removeRear(removing item from the bottom of the queue). You are given only stacks to implement this data structure. You can implement only push and pop operations. What are the total number of stacks required for this operation?(you can reuse the stack)

Select one:

- a. 3
- b. 2
- c. 4
- d. 1

Feedback

The correct answer is: 2

Question 21

59. A Javascript popup box used to alert dialog displaying the text "Welcome JS" is created by calling _____.

Select one:

- a. `window( "<alert> Welcome JS </alert>" );`
- b. `alert.window( "Welcome JS" );`
- c. `window.alert( "Welcome JS" );`
- d. `window( "alert = \"Welcome JS\"" );`

Feedback

The correct answer is: `window.alert( "Welcome JS" );`

Question 22

73. The data structure required to check whether an expression contains a balanced parenthesis is?

Select one:

- a. Queue
- b. Array
- c. Stack
- d. Tree

Feedback

The correct answer is: Stack

Question 23

86. Identify one which is preemption based algorithm when scheduling CPU:

Select one:

- a. FCFS

b. Random

c. Priority

d. SJF

Feedback

The correct answer is: Random

Question 24

55. _____ allow you to specify the style of your page elements (spacing, margins, etc.) separately from the structure of your document.

Select one:

a. Session

b. Style declarations

c. Cascading Style Sheets

d. Text Editors

Feedback

The correct answer is: Cascading Style Sheets

Question 25

36. You create some new tables and populate them with copies of your production data. You run reports on your new tables but your queries take a lot longer than on the original tables. What should you create to help speed up reads in the database?

Select one:

a. Index

b. Default

c. Stored procedure

d. Primary key

Feedback

The correct answer is: Index

Question 26

87. Which one of the following is NOT a mechanism of system threat that creates an environment to attack when operating system resources/ user files are misused?

Select one:

- a. Trojan
- b. denial of service
- c. port scanning
- d. Worm

Feedback

The correct answer is: Trojan

Question 27

20. Operator precedence of regular expression in decreasing order is

Select one:

- a. + , - , *
- b. * , + , -
- c. - , * , +
- d. - , + , *

Feedback

The correct answer is: * , + , -

Question 28

58. Which of the following is not correct about PHP?

Select one:

- a. Restrict users to access some pages of your website
- b. Encrypt and decrypt data using built-in algorithms
- c. Enable and disable cookies using setCookie() function
- d. Handle sessions of pages using setCookie() function

Feedback

The correct answer is: Handle sessions of pages using setCookie() function

Question 29

61. Which types of system testing uses a procedure that demands resources in abnormal quantity, frequency or volume?

Select one:

- a. Performance testing

- b. Stress testing
- c. Security testing
- d. Unit testing

Feedback

The correct answer is: Stress testing

Question 30

2. Type checking is one of the tasks performed by a compiler during

Select one:

- a. Syntax analysis
- b. Semantic analysis
- c. Intermediate code generation
- d. Lexical analysis

Feedback

The correct answer is: Semantic analysis

Question 31

Consider the following grammar,

$$E \rightarrow E+T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow \text{id} \mid (E)$$

which of the following statements are true about the parse tree constructed by deriving the string id+id*id using left-most derivation?

Select one:

- a. The tree contains 3 terminal nodes
- b. The height of the tree is 4
- c. The tree contains five internal nodes
- d. The root node of the tree contains T

Feedback

The correct answer is: The height of the tree is 4

Question 32

Suppose that you are given the following code snippet. Which one of the following statements is wrong about the given code snippet?

```
abstract class Animal {  
    public Animal (String name) {  
        System.out.println("I am animal class Constructor")  
    }  
    abstract void sound ();  
    final abstract void eat ();  
    final void move () {  
        //TODO  
    }  
}  
public class Lion extends Animal {  
    void sound () {  
        // TODO  
    }  
    public static void main (String [] args) {  
        // TODO  
    }  
}
```

Select one:

- a. It is possible to override the move () method in Lion class.
- b. The eat() method defined in Animal class is not supported
- c. It is must for the subclass to have at least one constructor.
- d. The sound () method in the super class must be implemented in the subclass Lion

Feedback

The correct answer is: It is possible to override the move () method in Lion class.

Question 33

16. What is the first step in data processing life cycle?

Select one:

- a. Testing
- b. Data acquisition
- c. Modeling
- d. Data representation

Feedback

The correct answer is: Data acquisition

Question 34

22. Regular expressions are

Select one:

- a. Type 3 language
- b. Type 2 language
- c. Type 1 language
- d. Type 0 language

Feedback

The correct answer is: Type 3 language

Question 35

23. Which of the following act violates cyber security?

Select one:

- a. Exploit
- b. Threat
- c. Attack
- d. Vulnerability

Feedback

The correct answer is: Attack

Question 36

92. Consider a banking application that needs to store customer information such as name, address, and account balance. A Customer class that contains private fields for these attributes is created, as well as getter and setter methods to access and modify those fields. This allows us to keep the data private and enforce business logic within the class, while still allowing other parts of the program to interact with the

object through well-defined interfaces. Which one of the following object-oriented concepts will fit with this application?

Select one:

- a. Inheritance
- b. Abstraction
- c. Encapsulation
- d. Polymorphism

Feedback

The correct answer is: Encapsulation

Question 37

75. The post-order traversal of a binary tree is O P Q R S T. Then possible pre-order traversal will be _____

Select one:

- a. T Q O S P R
- b. T Q O P S R
- c. T Q R S O P
- d. T O Q R P S

Feedback

The correct answer is: T Q O P S R

Question 38

82. Which of the following is a primary function of a domain controller in an Active Directory environment?

Select one:

- a. File sharing
- b. User management
- c. Authentication
- d. Searching files

Feedback

The correct answer is: Authentication

Question 39

78. _____ declares the host that's the most and best source of authoritative and DNS information for the zone.

Select one:

- a. DNS A Record
- b. DNS NS Record
- c. DNS PTR Record
- d. DNS Start of Authority Record

Feedback

The correct answer is: DNS Start of Authority Record

Question 40

91. Which of the following is a correct syntax to create an object of interface named Animal?

Select one:

- a. It is not supported to create an object of Animal
- b. new Animal ();
- c. Animal animal = new Animal ();
- d. Interface Animal animal = new Animal ();

Feedback

The correct answer is: It is not supported to create an object of Animal

Question 41

27. Which of the following is not a cybercrime?

Select one:

- a. Man in the Middle
- b. AES
- c. Malware
- d. Denial of Service

Feedback

The correct answer is: AES

Question 42

41. Suppose you are an ORACLE DBA and you need to create a trigger on the EMPLOYEES table that monitors every row that is changed and places this information into the AUDIT_TABLE. What type of trigger do you create?

Select one:

- a. Statement-level trigger on the AUDIT_TABLE table.
- b. FOR EACH ROW trigger on the AUDIT_TABLE table.
- c. Statement-level trigger on the EMPLOYEES table.
- d. FOR EACH ROW trigger on the EMPLOYEES table.

Feedback

The correct answer is: FOR EACH ROW trigger on the EMPLOYEES table.

Question 43

50. Which is the correct statement about operator overloading in C++?

Select one:

- a. Only arithmetic operators can be overloaded
- b. Only non-arithmetic operators can be overloaded
- c. Precedence of operators are changed after overloading
- d. Associativity and precedence of operators does not change

Feedback

The correct answer is: Associativity and precedence of operators does not change

Question 44

52. An/A _____ is ideal when the style is applied to many pages.

Select one:

- a. external style sheet
- b. style sheet
- c. internal style sheet
- d. cascading style sheet

Feedback

The correct answer is: external style sheet

Question 45

42. What is the best data type definition for Oracle when a field is alphanumeric and has a fixed length?

Select one:

- a. NUMBER
- b. CHAR
- c. LONG
- d. VARCHAR2

Feedback

The correct answer is: CHAR

Question 46

26. Which of the following is the least strong security encryption standard in WLAN?

Select one:

- a. WPA3
- b. WEP
- c. WPA
- d. WPA2

Feedback

The correct answer is: WEP

Question 47

64. Polymorphism reduces the effort required to extend an object system by

Select one:

- a. Coupling objects together more tightly
- b. Removing the barriers imposed by encapsulation
- c. Making objects more dependent on one another
- d. Enabling a number of different operations to share the same name

Feedback

The correct answer is: Enabling a number of different operations to share the same name

Question 48

53. Which of the following task cannot be done by JavaScript?

Select one:

- a. to manipulate HTML elements
- b. Handle to events, errors and exceptions
- c. to access database data and file system
- d. to validate data

Feedback

The correct answer is: to access database data and file system

Question 49

21. A language is regular if and only if

Select one:

- a. accepted by LBA
- b. accepted by DFA
- c. accepted by PDA
- d. accepted by Turing machine

Feedback

The correct answer is: accepted by DFA

Question 50

84. Sector 0 of a disk which is used to boot the computer is called_____.

Select one:

- a. MBR
- b. Superblock
- c. Partition
- d. Bootblock

Feedback

The correct answer is: MBR

Question 51

12. Depth first search always expands the _____ node in the problem representation of a search tree.

Select one:

- a. Shallowest
- b. Minimum cost

- c. Deepest
- d. Child node

Feedback

The correct answer is: Deepest

Question 52

81. Which of the following backup technique is most space efficient?

Select one:

- a. Incremental backup
- b. RAID Technology
- c. Differential backup
- d. Full backup

Feedback

The correct answer is: Incremental backup

Question 53

85. Which operating system security is used to identify each user of the system and associate the executing programs with those users?

Select one:

- a. Confidentiality
- b. Separation
- c. Authentication
- d. Integrity

Feedback

The correct answer is: Authentication

Question 54

24. Which of the following is an internet scam done by cyber-criminals where the user is convinced digitally to provide confidential information?

Select one:

- a. DoS attack
- b. MiTM attack

- c. Website attack
- d. Phishing attack

Feedback

The correct answer is: Phishing attack

Question 55

3. Which one of the following is not true about stack-based shift-reduce parser?

Select one:

- a. The parser repeats shifting and reducing actions until the stack is empty or an error is happened
- b. Initially the stack contains \$ and starting none terminal of the grammar
- c. \$ is used to mark the bottom of the stack and the right end of the input buffer
- d. The parsing is successful if the stack is left with \$ and the starting symbol of the grammar and the input buffer is left with \$

Feedback

The correct answer is: Initially the stack contains \$ and starting none terminal of the grammar

Question 56

31. One is not correct about binary number system.

Select one:

- a. it is also used as a machine language
- b. It is positional weighted number
- c. The base is 3
- d. The symbols used are 0 and 1

Feedback

The correct answer is: The base is 3

Question 57

1. One of the following is true about lexical and syntax analyzers

Select one:

- a. Lexical analyzer works on simple recursive constructs a language
- b. Scanner is a co-routine of parser
- c. Syntax analyzer is a co-routine of lexical analyzer

d. Syntax analyzer works on simple none recursive constructs a language

Feedback

The correct answer is: Scanner is a co-routine of parser

Question 58

8. Which one of the following applications of network is used to assign IP address to hosts automatically?

Select one:

a. File transfer protocol

b. Mailing system

c. DNS

d. DHCP

Feedback

The correct answer is: DHCP

Question 59

69. Among the following set of problems listed below, identify the one which can be solved using divide and conquer strategy?

Select one:

a. N-queens problem.

b. Graph coloring problem

c. Finding the maximum continuous sum in the array.

d. Minimum spanning tree problem.

Feedback

The correct answer is: Finding the maximum continuous sum in the array.

Question 60

How can the index.php script access the email form element of the following HTML form?

```
<form action="index.php" method="post">  
  <input type="text" name="email"/> </form>
```

Select one:

a. \$_POST['text']

- b. `$_GET['email']`
- c. `$_SESSION['text']`
- d. `$_POST['email']`

Feedback

The correct answer is: `$_POST['email']`

Question 61

25. Which of the following DDoS in mobile systems wait for the owner to trigger the cyber attack?

Select one:

- a. Programs
- b. Virus
- c. Botnets
- d. Worms

Feedback

The correct answer is: Botnets

Question 62

34. Which is a representation of information transfer from one register to another in symbolic form?

Select one:

- a. `MOV R1,R2`
- b. `R1->R2`
- c. `R1=R2`
- d. `SH R1,R2`

Feedback

The correct answer is: `MOV R1,R2`

Question 63

The following java code snippet shows a class definition containing only parametrized constructor. Based on the given code segment below, which of the following statement wrongly invokes the constructor?

```
public class Car {
```

```
    String type;
```

```
String model;  
  
String color;  
  
public Car(String type, String model, String color) {  
  
    this.type = type;  
  
    this.model = model;  
  
    this.color = color;  
  
}  
  
}
```

Select one:

- a.
Car auris = new Car ("Toyota", "Auris");
- b.
new Car ("Volkswagen", "Golf", "green");
- c.
Car golf = new Car ("Volkswagen", "Golf", "green");
- d.
Car focus = new Car ("Ford", "Focus", "red");

Feedback

The correct answer is:

Car auris = new Car ("Toyota", "Auris");

Question 64

5. Which of the following concept of FSA is used in the compiler?

Select one:

- a. Code generation

- b. Lexical analysis
- c. Parser
- d. Code optimization

Feedback

The correct answer is: Lexical analysis

Question 65

15. Which one of the following is different?

Select one:

- a. DFS
- b. Informed Searching
- c. Uninformed searching techniques
- d. BFS

Feedback

The correct answer is: DFS

Question 66

17. The action of the Simple reflex agent completely depends upon _____.

Select one:

- a. Perception history
- b. Utility functions
- c. Current perception
- d. Learning theory

Feedback

The correct answer is: Learning theory

Question 67

The expression `static_cast(6.9) + static_cast(7.9)` evaluates to ____.

Select one:

- a. 13
- b. 14.8
- c. 14

d. 15

Feedback

The correct answer is: 13

Question 68

Given the following set of unbreakable items with their associated values in Birr and weights in kilograms in a table, what is the maximum profit that can be made having a 0/1 knapsack with a capacity of 6 kg.

Value(Birr.)	60	20	32	15	40	36
Weight(Kg.)	4	2	2	1	3	2

Select one:

a. 98Br

b. 88Br

c. 96Br

d.

86Br

Feedback

The correct answer is: 96Br

Question 69

57. A web based application model that is used to add, modify and access objects hieratically.

Select one:

a. RPC

b. MVC

c. RMI

d. DOM

Feedback

The correct answer is: DOM

Question 70

71. A linear collection of data elements where the linear node is given by means of pointer is called?

Select one:

- a. Linked list
- b. Node list
- c. Primitive list
- d. Unordered list

Feedback

The correct answer is: Linked list

Question 71

43. In which database schema user views are included?

Select one:

- a. Internal
- b. Physical
- c. External
- d. Conceptual

Feedback

The correct answer is: External

Question 72

89. Which one of the following is not contained in a process stack?

Select one:

- a. PID of child process
- b. Return addresses
- c. Local variables
- d. Function parameters

Feedback

The correct answer is: PID of child process

Question 73

63. Which of the following is not an objective of use-case modeling?

Select one:

- a. To provide a description of end user and system interaction
- b. To define the functional and operational requirements

- c. To provide a basis for validation testing
- d. To define the hierarchy for the system

Feedback

The correct answer is: To define the hierarchy for the system

Question 74

77. . Which of the following is an infix expression?

Select one:

- a. +ab
- b. abc+*
- c. (a+b)*(c+d)
- d. ab+c*

Feedback

The correct answer is: (a+b)*(c+d)

Question 75

18. If we have more than one accepting states or an accepting state with an outdegree, which of the following actions will be taken?

Select one:

- a. addition of new state
- b. make the newly added state as final
- c. more than one option is correct
- d. removal of a state

Feedback

The correct answer is: more than one option is correct

Question 76

49. What is an inline function?

Select one:

- a. A function that is not checked for syntax errors
- b. A function that is not checked for semantic analysis
- c. A function that is expanded at each call during execution

d. A function that is called during compile time

Feedback

The correct answer is: A function that is expanded at each call during execution

Question 77

44. _____ limits who gains access to the database while _____ limits what a user can access within the database.

Select one:

- a. Access authentication, view definition
- b. Access control, database security
- c. Access authentication, user definition
- d. Data access, user monitoring

Feedback

The correct answer is: Access authentication, view definition

Question 78

10. Which of the following is true about MAC address?

Select one:

- a. It is used for subnetting purpose
- b. It is represented by 128 bits
- c. It is also known as logical address
- d. It is commonly assigned by the manufacturer of the NIC

Feedback

The correct answer is: It is commonly assigned by the manufacturer of the NIC

Question 79

11. Which of the following devices is layer 3(network Layer) device?

Select one:

- a. Router
- b. Bridge
- c. Hub
- d. Switch

Feedback

The correct answer is: Router

Question 80

74. Which of the following is true about arrays?

Select one:

- a. Easier to store elements of same data type
- b. Elements in an array cannot be sorted
- c. Index of first element of an array is 1
- d. Objects of mixed data types can be stored

Feedback

The correct answer is: Easier to store elements of same data type

Question 81

51. All of the following are examples of integral data types EXCEPT _____.

Select one:

- a. int
- b. short
- c. char
- d. double

Feedback

The correct answer is: double

Question 82

38. What is the impact of not including a JOIN command when using multiple tables in a query?

Select one:

- a. All records in each table are associated with all records in the other tables
- b. The foreign keys in each table are linked to the primary keys in the other tables
- c. The primary keys in each table are joined together
- d. The query will not work – an error is generated

Feedback

The correct answer is: All records in each table are associated with all records in the other tables

Question 83

29. The fundamental building block of all digital logic circuits is _____.

Select one:

- a. Adder
- b. Circuit
- c. Data
- d. Gates

Feedback

The correct answer is: Gates

Question 84

68. The amortized time complexity of the dynamic array problem is?

Select one:

- a. $O(1)$
- b. $O(n \log n)$
- c. $O(\log n)$
- d. $O(n)$

Feedback

The correct answer is: $O(1)$

Question 85

56. _____ define the web contents and their structures, _____ manage the layout of the web contents and their elements and _____ control the interactivity of web site pages respectively.

Select one:

- a. Javascript, VBscript and PHP
- b. HTML, CSS and PHP
- c. HTML, CSS and XLS
- d. CSS, HTML, XSL

Feedback

The correct answer is: HTML, CSS and PHP

Question 86

$(a+b)^*$ is equivalent to

Select one:

- a. b^*a^*
- b. a^*b^+
- c. $(a^*b^*)^*$
- d. a^*b^*

Feedback

The correct answer is: b^*a^*

Question 87

67. Which one of the following is not the advantage of using randomized algorithms?

Select one:

- a. Simplicity of implementation.
- b. Achieving worst case time-complexity.
- c. Implementing problems requiring randomization.
- d. Efficiency in running time.

Feedback

The correct answer is: Achieving worst case time-complexity.

Question 88

37. Suppose there is a database that maintains business activities of the company. In the database Sales data is stored into two tables—one for recent sales (the past six months) and one for older or archived sales data. You are tasked with creating a solution to allow managers to be able to see monthly sales data from the previous month and for the same period a year ago. What would you do?

Select one:

- a. Create indexes on the two tables based on the date and query the tables individually.
- b. Combine the data into a single table and query it.
- c. Create a trigger to identify when data is added and query the new data.
- d. Create a view that joins data from both tables.

Feedback

The correct answer is: Create a view that joins data from both tables.

Question 89

62. Which system development method executes the phases of the SDLC in linear manner?

Select one:

- a. Scrum
- b. Prototyping
- c. XP
- d. Waterfall model

Feedback

The correct answer is: Waterfall model

Question 90

90. Which one of the following is wrong about interface?

Select one:

- a. Interface is used to achieve full abstraction, loose coupling and multiple inheritance
- b. The default variable modifiers in an interface are public, static and final
- c. An interface can create an object of its own
- d. Methods in an interface are abstract, and public by default

Feedback

The correct answer is: An interface can create an object of its own

Question 91

47. Which one of the following is a non-linear dynamic data structure?

Select one:

- a. Tree
- b. Queue
- c. Array
- d. Stack

Feedback

The correct answer is: Tree

Question 92

30. The structure and behavior of various functional modules (HW and SW) of a digital computer as well as how these modules interact to meet the requirement of the user is_____.

Select one:

- a. Computer organization
- b. Computer architecture
- c. computer specification
- d. computer application

Feedback

The correct answer is: Computer architecture

Question 93

28. Algorithms are designed to attain the following goal:

Select one:

- a. to save time and memory resources
- b. to keep personal information private
- c. to solve some common problems in our daily life
- d. to manage users in computer system

Feedback

The correct answer is: to save time and memory resources

Question 94

32. The most basic digital arithmetic circuit which Performs the addition of two binary digits.

Select one:

- a. Full adder
- b. Half adder
- c. decoder
- d. multiplexer

Feedback

The correct answer is: Half adder

Question 95

What is the output of the following java program?

```

class Flower {
    public Flower () {
        this ("Rose ");
        System.out.print("Flower. ");
    }
    public Flower (String name) {
        System.out.print("I am "+name);
    }
}
public class RoseTest extends Flower {
    public RoseTest (){
        super ();
        System.out.print("Hi there! ");
    }
    public static void main(String [] args) {
        Flower flower = new RoseTest();
    }
}

```

Select one:

- a. Hi there! Rose flower I am.
- b. Hi there! I am Rose Flower.
- c. I am Flower. Rose! Hi there!
- d. I am Rose Flower. Hi there!

Feedback

Your answer is incorrect.

The correct answer is: I am Rose Flower. Hi there!

Question 96

33. A register that hold the memory address of the instruction to be executed next is _____.

Select one:

- a. Accumulator
- b. data register
- c. Program counter
- d. Address register

Feedback

The correct answer is: Program counter

Question 97

48. Which of the following is true about FILE *fp?

Select one:

- a. FILE is a keyword in C++ for representing files and fp is a variable of FILE type.
- b. FILE is a buffered stream
- c. FILE is a structure and fp is a pointer to the structure of FILE type
- d. FILE is a stream

Feedback

The correct answer is: FILE is a structure and fp is a pointer to the structure of FILE type

Question 98

35. Amongst the ACID properties of a transaction, the 'Durability' property requires that the changes made to the database by a successful transaction persist

Select one:

- a. Except in case of a power failure
- b. Except in case of an operating system crash
- c. Except in case of a disk crash
- d. Always, even if there is a failure of any kind

Feedback

The correct answer is: Always, even if there is a failure of any kind

Question 99

7. What kind of transmission medium is most appropriate to carry data in a computer network that is exposed to electrical interferences?

Select one:

- a. Unshielded twisted pair
- b. Optical fiber
- c. Microwave
- d. Coaxial cable

Feedback

The correct answer is: Coaxial cable

Question 100

72. The postfix form of the expression $(A + B) * (C * D - E) * F / G$ is?

Select one:

- a. $AB + CD * E - F ** G /$
- b. $AB + CD * E - * F * G /$
- c. $AB + CD * E - FG /**$
- d. $AB + CDE * - * F * G /$

Feedback

The correct answer is: $AB + CD * E - * F * G /$